

# Dae Yeon Kim Ph.D.

Assistant Professor, Chemical and Biomolecular Engineering, Chonnam National University

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## EDUCATION

|              |                                                                                                                                                                                                                                                                     |          |
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| <b>Ph.D.</b> | <b>Seoul National University</b> , Seoul, Korea<br>School of Chemical and Biological Engineering<br><br>Thesis: “A study on particle deposition and clogging dynamics of colloidal suspensions in complex flow”<br>(Advisor: Prof. <a href="#">Kyung Hyun Ahn</a> ) | Aug 2023 |
| <b>B.S.</b>  | <b>Sogang University</b> , Seoul, Korea<br>Department of Chemical and Biomolecular Engineering<br><br>Graduation honors, <i>Summa Cum Laude</i>                                                                                                                     | Feb 2017 |

## PROFESSIONAL EXPERIENCES

|                                |                                                                                                                                                         |                     |
|--------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|
| <b>Assistant Professor</b>     | Department of Chemical and Biomolecular Engineering<br>Chonnam National University, Yeosu-si, Jeollanam-do, Korea                                       | Mar 2026 – Present  |
| <b>Postdoctoral Scholar</b>    | Department of Chemical Engineering<br>Stanford University, Stanford, CA, USA<br>(Advisor: Prof. <a href="#">Sho Takatori</a> )                          | Feb 2025 – Jan 2026 |
| <b>Postdoctoral Scholar</b>    | Department of Chemical Engineering<br>University of California, Santa Barbara, Santa Barbara, CA, USA<br>(Advisor: Prof. <a href="#">Sho Takatori</a> ) | Mar 2024 – Jan 2025 |
| <b>Postdoctoral Researcher</b> | Center for Coating Materials and Processing<br>Seoul National University, Seoul, Korea<br>(Advisor: Prof. <a href="#">Kyung Hyun Ahn</a> )              | Sep 2023 – Jan 2024 |

## RESEARCH EXPERIENCES

### Research Skills

|                     |                                                                                                                                                                                                                                     |
|---------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Microfluidics       | <ul style="list-style-type: none"><li>• Design and fabrication of microchannel</li><li>• Fluid visualization and PIV</li><li>• Pressure drop measurement</li></ul>                                                                  |
| Rheometry           | <ul style="list-style-type: none"><li>• Rotational rheometer (ARES, AR-G2, DHR)</li><li>• Extensional rheometer (Capillary Breakup Extensional Rheometer)</li><li>• Passive/Active microrheology (Optical tweezer)</li></ul>        |
| Capillary breakup   | <ul style="list-style-type: none"><li>• Measurement of elongational property of complex fluids (polymer solution, colloidal suspension)</li><li>• Direct visualization of thinning filament surface via high-speed camera</li></ul> |
| Optical tweezer     | <ul style="list-style-type: none"><li>• Measurement of inter-particle interaction (e.g. Hydrodynamic interaction)</li><li>• Microrheology of soft matter</li></ul>                                                                  |
| Chemical labeling   | <ul style="list-style-type: none"><li>• Soft vesicle labeling (SUV, GUV)</li><li>• Colloids and bacteria labeling</li></ul>                                                                                                         |
| Bacterial culturing | <ul style="list-style-type: none"><li>• Characterization and labeling</li><li>• Quantitative imaging analysis</li></ul>                                                                                                             |

- Analysis tool
- Image processing and quantitative analysis (MATLAB, Python)
  - CFD simulation (COMSOL)

## Research Interests

Complex materials and soft matter physics

- Colloidal suspension rheology
- Fluid-mediated interactions between colloids
- Active matter rheology and control
- Colloidal fouling onto a membrane surface (complex interactions between foulants and membrane)
- Flow instability of viscoelastic fluid

Heterogeneity control of industrial slurry (e.g. lithium-ion batteries, fuel cells, CMP slurry)

- Stability control of colloidal slurry
- Characterization of slurry for quality control

Data-driven machine learning classification

- Characterization of complex fluid
- Clogging and fouling prediction

Image-based rheological properties measurement

- Discovery of constitutive relations for complex fluids
- Dynamics of soft materials with active forcing

## PUBLICATIONS

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### Peer-reviewed Journal Articles

\* equal contribution    † co-corresponding

13. Haegyo Jeong, Minsun Hwang, **Dae Yeon Kim**<sup>†</sup>, Seon Yeop Jung<sup>†</sup>, Influence of fluid viscosity and cavity length on particle transport and deposition in expansion-contraction microchannels, *Korea-Australia Rheology Journal*, **2026**, 38, 335–354. [\[Link\]](#)
12. **Dae Yeon Kim**<sup>\*</sup>, Sachit G. Nagella<sup>\*</sup>, Saksham Malik, Nayeon Park, Jaewook Nam, Eric S.G. Shaqfeh, Sho C. Takatori, Colloidal hydrodynamic interactions in viscoelastic fluids, *Soft Matter*, **2025**, 21, 8035–8048. [\[Link\]](#)
11. Minsun Hwang, **Dae Yeon Kim**, Kyung Hyun Ahn, Sung Sik Lee, Seon Yeop Jung, Particle deposition in colloid-polymer suspensions: Complex interplay between flow characteristics and polymer interactions, *Physics of Fluids*, **2025**, 37, 073389. [\[Link\]](#)
10. **Dae Yeon Kim**<sup>\*</sup>, Sachit G. Nagella<sup>\*</sup>, Kyu Hwan Choi, Sho C. Takatori, Direct experimental measurement of many-body hydrodynamic interactions with optical tweezers, *Physical Review Fluids*, **2025**, 10, 064301. [\[Link\]](#)
9. Woobin Yi<sup>\*</sup>, **Dae Yeon Kim**<sup>\*</sup>, Howon Jin, Sangwoong Yoon, Kyung Hyun Ahn, Early detection of pore clogging in microfluidic systems with 3D convolutional neural network, *Separation and Purification Technology*, **2025**, 359, 130428. [\[Link\]](#)
8. Young Jin Lee, **Dae Yeon Kim**, Kyung Hyun Ahn, Particulate fouling simulation in unit micropore using a hydrodynamically coupled Lagrangian framework, *Desalination*, **2025**, 593, 118255. [\[Link\]](#)
7. Minsun Hwang, **Dae Yeon Kim**, Seon Yeop Jung, Kyung Hyun Ahn, Effect of fluid viscosity on the deposition of colloidal particles in an expansion-contraction microchannel flow, *Nihon Reorogi Gakkaishi*, **2024**, 52 (4), 231–240. [\[Link\]](#)

6. Wonsik Shin, Joung Sook Hong, **Dae Yeon Kim**, Si Yoon Kim, Kyu Hyun, Jun Dong Park, Kyung Hyun Ahn, Effect of cellulose nanocrystals on the emulsion stability and rheological properties of microalgal Pickering emulsions, *Algal Research*, **2024**, 83, 103731. [\[Link\]](#)
5. Young Jin Lee, Howon Jin, **Dae Yeon Kim**, Seunghoon Kang, Kyung Hyun Ahn, Lagrangian-based simulation method using constrained Stokesian dynamics for particulate flows in microchannel, *Journal of Fluid Mechanics*, **2024**, 990, A3. [\[Link\]](#)
4. **Dae Yeon Kim**, Seon Yeop Jung, Young Jin Lee, Howon Jin, Kyung Hyun Ahn, From successive deposition to clogging of poly(styrene) particles in the cross-flow through a T-shaped microchannel, *Separation and Purification Technology*, **2023**, 322, 124263. [\[Link\]](#)
3. **Dae Yeon Kim**, Seon Yeop Jung, Young Jin Lee, Kyung Hyun Ahn, Effect of colloidal interactions and hydrodynamic stress on particle deposition in a single micropore, *Langmuir*, **2022**, 38 (19), 6013–6022. [\[Link\]](#)
2. **Dae Yeon Kim**, Kyung Hyun Ahn, Effect of operating conditions on the sessile drop oscillation of PEO solution in the capillary thinning process, *Nihon Reorogi Gakkaishi*, **2021**, 49 (2), 127–133. [\[Link\]](#)
1. Youngseok Kim, **Dae Yeon Kim**, Joung Sook Hong, Kyung Hyun Ahn, Agglomerate breakup of destabilized polystyrene particles under a cross-channel planar extensional flow, *Langmuir*, **2018**, 34 (38), 11454–11463. [\[Link\]](#)

### Submitted/in-progress

3. **Dae Yeon Kim**, Free-vibration mechanism and viscoelastic damping of sessile-drop oscillation in capillary thinning, *in preparation*.
2. **Dae Yeon Kim**, Colloidal hydrodynamic interactions: from bulk rheometry to rotation-based optical tweezer measurements, *under review*.
1. **Dae Yeon Kim**, Izzie Allydice, Sho C. Takatori, Colloidal hydrodynamic interaction in bacterial active bath, *in preparation*.

## PRESENTATIONS

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### International Conference

9. Fluid-mediated rotational coupling and hydrodynamic interactions in colloidal particles using optical trapping, *APS Global Physics Summit (APS March Meeting 2025)*. 16 – 21 Mar 2025, Anaheim, CA, USA.
8. Successive deposition of poly(styrene) particles in cross-flow through T-shaped microchannel, *The 19th International Congress on Rheology (ICR 2023)*. 29 July – 4 Aug 2023, Athens, Greece.
7. Effect of salt concentration and flow rate on clogging dynamics at the single micro-pore. *The 8th Pacific Rim Conference on Rheology (PRCR 2023)*. 15 – 19 May 2023, Vancouver, BC, Canada.
6. Effect of salt concentration and flow rate on clogging dynamics at the single micro-pore. *The 93rd Annual Meeting of the Society of Rheology (SoR 93rd Annual Meeting)*. 9 – 13 Oct 2022, Chicago, IL, USA.
5. Effect of salt concentration and flow rate on clogging dynamics at the single micro-pore. *11th Korea-Australia Rheology Conference (KARC 2021)*. 24 – 25 Nov 2021, Virtual.

4. The effect of operating condition and rheological properties of PEO solution on the sessile drop oscillation during elasto-capillary thinning. *International Union of Materials Research Societies-International Conference in Asia (IUMRS-ICA 2020)*. 23 – 26 Feb 2021, Virtual.
3. Effect of rheological properties of PEO solution on the sessile drop oscillation during the early stages of elasto-capillary thinning transition. *The 15th International Workshop for East Asian Young Rheologists (IWEAYR 2020)*. 8 – 10 Jan 2020, Changchun, China.
2. Effect of rheological properties of PEO solution on the sessile drop oscillation during the early stages of elasto-capillary thinning transition. *Annual European Rheology Conference (AERC 2019)*. 8 – 11 Apr 2019, Portoroz, Slovenia.
1. Agglomerate breakup of destabilized polystyrene particles under a cross-channel planar extensional flow. *The 14th International Workshop for East Asian Young Rheologists (IWEAYR 2019)*. 23 – 26 Jan 2019, Nagoya, Japan.

### Others (2023 – Present)

4. Probing hydrodynamic interactions and rheology in complex fluids using optical laser tweezers. *2026 Annual Meeting of the Korean Society of Rheology*. 7 – 8 May 2026, Seoul, Korea
3. Prediction of membrane fouling via particle transport phenomena and ai-based analysis. *2026 Spring Conference of the Korean Society for Energy*. 15 – 17 Apr 2026, Yeosu, Korea
2. Direct experimental measurement of hydrodynamic interactions using optical tweezers. *Shaqfeh Group Seminar, Stanford University*. May 2025
1. A study on particle deposition and clogging dynamics of colloidal suspensions in complex flow. *EEPL Group Seminar, Dankook University*. Aug 2023

## TEACHING AND MENTORING

|                                       |                                                                                       |                        |
|---------------------------------------|---------------------------------------------------------------------------------------|------------------------|
| <b>Teaching</b><br>(S:Spring, F:Fall) | General Chemistry I (Chonnam National University)                                     | 2026S                  |
|                                       | Chemical Reaction Engineering (Chonnam National University)                           | 2026S                  |
|                                       | Creative Academy: Basic Chemical Engineering<br>(Janghoon High school, Seoul)         | Mar 2018 –<br>Jan 2019 |
| <b>TA</b>                             | Transport Phenomena (Seoul National University)                                       | Spring. 2018           |
|                                       | Introduction to Chemical Engineering (Seoul National University)                      | Fall. 2019             |
| <b>Mentoring</b>                      | 2025 ChemE REU Program                                                                | Jul 2025 –             |
|                                       | Mentee: Izzie Allydice (Stanford '28)                                                 | Present                |
|                                       | EEPL Group, Dankook University<br>Mentee: Minsun Hwang (B.S.–M.S. Integrated Program) | Mar 2022 –<br>Jul 2025 |

## AWARDS AND HONORS

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|--------------------------------|--------------------------------------------------------------------------------------------------------------|------------------------|
| <b>Best Paper Award</b>        | DONGJIN Best Paper Award<br>(supported by DONGJIN SEMICHEM CO., LTD.)                                        | Jul 2022               |
| <b>Best Presentation Award</b> | Best Presentation Award of 15th IWEAYR<br>(The 15th International Workshop for East Asian Young Rheologists) | Jan 2020               |
| <b>Academic Scholarship</b>    | Scholarship for academic excellence<br>(Sogang University)                                                   | Sep 2013 –<br>Aug 2016 |

## OTHER ACTIVITIES

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|-----------------------|---------------------------------------------------------------------------------|---------------------|
| <b>Member</b>         | Young Engineers Honor Society (YEHS)                                            | Jan 2014 – Present  |
| <b>Student Leader</b> | Department of Chemical and Biomolecular Engineering<br>Sogang University, Korea | Jan 2014 – Mar 2014 |

## REFERENCES

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|---------------------------|---------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------|
| <b>Dr. Kyung Hyun Ahn</b> | Professor,<br>School of Chemical and Biological Engineering,<br>Seoul National University, Seoul, Korea | <a href="mailto:ahnnet@snu.ac.kr">ahnnet@snu.ac.kr</a>             |
| <b>Dr. Sho Takatori</b>   | Associate Professor,<br>Department of Chemical Engineering,<br>Stanford University, Stanford, CA        | <a href="mailto:stakatori@stanford.edu">stakatori@stanford.edu</a> |
| <b>Dr. Seon Yeop Jung</b> | Assistant Professor,<br>Department of Chemical Engineering,<br>Dankook University, Gyeonggi-do, Korea   | <a href="mailto:seon27@dankook.ac.kr">seon27@dankook.ac.kr</a>     |